

Pharmaceutical solutions

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Objectives

At the end of these topics student will be able to understand

- The different solvents and formulation additives used in solution dosage forms.
- The advantage and disadvantage of solution dosage forms
- The types of solution dosage forms and their use
- Methods of preparation of different solution dosage forms
- Methods used to speed up the dissolution of solute

Definition

- In pharmaceutical terms, **solutions** are **liquid preparations** that contain one or more drugs dissolved in a **suitable solvent** or mixture of **mutually miscible solvents**.

Formulation of solutions

Solution formulation have different constituents.

- API
- Solvent
- Excipients
 - ✓ Buffers
 - ✓ Isotonicity modifiers
 - ✓ Viscosity enhancement
 - ✓ Preservatives
 - ✓ Reducing agents and antioxidants
 - ✓ Sweetening agents
 - ✓ Colors, etc

Solvent

- Water is the solvent most widely used as a vehicle for pharmaceutical products.

Advantages of water:

- ✓ It possesses a high dielectric constant, which is essential for ensuring the dissolution of a wide range of ionizable materials
 - ✓ Cheap.
 - ✓ Tasteless,
 - ✓ Odorless,
 - ✓ Inert (lack of pharmacological activity)
 - ✓ Neutral
 - ✓ Physiologically compatible
 - ✓ Lack of toxicity

Solvent

Disadvantages of water

- Support growth of microorganisms when contaminated
- Favorable medium for many chemical reactions.
- Some hydrolysable drugs such as glycosides and esters form unstable solutions when dissolved in water.

Solvent...

- Different grades of water are used for different kind of solutions.
- **Water for Injections** must be used for the formulation of parenteral solutions
- Similarly, drugs which are liable to oxidation, such as apomorphine and ergotamine maleate, require **Water for Injections BP free from dissolved air** to be used.

Solvent

Approaches to improve aqueous solubility

- Although water is very widely used for inclusion in pharmaceutical preparations, it may not be possible to ensure complete solution of all ingredients at all normal storage temperatures.
- **Co solvency**
- **PH control**
- **Solubilization**
- **Complexation**
- **Chemical modification**
- **Particle size control**

Solvent....

Non aqueous solvents

- are used if it is not possible to ensure complete solution of the ingredients at all storage temperatures or
- if the drug is unstable in aqueous systems

The use of non-aqueous systems may also have other advantages.

- For example, the intramuscular injection of solutions of drugs in oils is often used for depot therapy, and
- some drugs are specifically synthesized to improve their oil solubilities.

Solvent....

Alcohols

- Ethyl alcohol is the most widely used solvent
- Particularly for external application
- It is also particularly useful for the extraction of crude drugs, being more selective than water.
- At concentrations greater than 15% ethanol exhibits antimicrobial activity,
- but because of its toxicity it is used orally or parenterally only at low concentrations, usually as a co solvent with water

Solvent.....

propylene glycol

- often used in conjunction with water or glycerol as a cosolvent.
- It is used, for example, in the formulation of Digoxin Injection, Phenobarbital Injection, and some formulations of Diazepam Injection, Co-trimoxazole Intravenous Infusion and
- as the diluent for both Chloramphenicol Ear Drops and some brands of hydrocortisone ear drops, and in many preparations for oral use.

Solvent

Ethyl ether

- This material is widely used for the extraction of crude drugs, but because of its own therapeutic activity **it is not used for the preparation of formulations for internal use.**
- It is, however, used as a co solvent with alcohol in some **collodions.**

Liquid paraffin

- it is often used as a solvent for the topical application of drugs in emulsion formulations

Formulation Additives

Buffers

- These are materials which, when dissolved in solvent, will enable the solution **to resist any change in pH** should an acid or an alkali be added.
- The choice of suitable buffer depends on the pH and buffering capacity required.
- It must be **compatible with other excipients** and have a **low toxicity**.
- As the pH of most body fluids is 7.4, products such as injections, eye drops and nasal drops should, in theory, be buffered at this value to avoid irritation.

Formulation Additives.....

Isotonicity modifiers

- Solutions for injection, for application to mucous membranes, and large-volume solutions for ophthalmic use must be made iso-osmotic with tissue fluid to **avoid pain and irritation**.
- The most widely used isotonicity modifiers are **dextrose and sodium chloride**.

Preservatives

When choosing a suitable preservative it must be ensured that:

- **adsorption of the preservative onto the container from the product does not occur, and**
- **its efficiency is not impaired by the pH of the solution or by interactions with other ingredients**

Formulation Additives.....

Viscosity enhancement

- It may be difficult for aqueous-based topical solutions to remain in place on the **skin or in the eyes** for any significant time because of their low viscosities.
- To counteract this effect, low concentrations of **gelling agents** can be used to increase the apparent viscosity of the product.
- Examples include povidone, hydroxyethylcellulose and carbomer.

Formulation Additives.....

Reducing agents and antioxidants

The decomposition of pharmaceutical products by oxidation can be controlled by

- ❑ The addition of reducing agents such as sodium metabisulphite, or
- ❑ Antioxidants such as butylated hydroxyanisole or butylated hydroxytoluene.

Formulation Additives.....

Sweetening agents

- Low molecular weight carbohydrates, and in particular sucrose, are traditionally the most widely used sweetening agents.
- Sucrose has the advantage of being color less, very soluble in water, stable over a pH range of about 4-8 and,
- by increasing the viscosity of fluid preparations, will impart to them a pleasant texture in the mouth.

Formulation Additives.....

- It will mask the tastes of both salty and bitter drugs and has a soothing effect on the membranes of the throat.
- For this reason, despite its cariogenic properties, sucrose is particularly useful as a vehicle for **antitussive preparations**.
- **Polyhydric alcohols such as sorbitol**, mannitol and, to a lesser extent glycerol, also possess sweetening power and can be included in preparations **for diabetic use, where sucrose is undesirable**.

Formulation Additives.....

Flavours and perfumes

- This is particularly useful in paediatric formulation to ensure **patient compliance**.
- The inclusion of flavours has the additional advantage of enabling the easy identification of liquid products.
- Flavouring and perfuming agents can be obtained from either natural or synthetic sources.
- Natural products include **fruit juices, aromatic oils such as peppermint and lemon can be used**

Formulation Additives.....

Colors

- Are used in order to improve the attractiveness of the product.
- **To enable easy product identification**

Advantages of solution dosage forms :

- ✓ Easier to swallow than solids
- ✓ particularly acceptable for **pediatrics and geriatrics**
- ✓ **Onset of drug action is fast from solutions than solid dosage forms**
- ✓ Minimized irritation effect of some drugs(e.g., aspirin, KCl, etc)

Disadvantages of solution dosage forms:

- ✓ Bulky and inconvenient to transport and store
- ✓ **The stability of ingredients in aqueous solution is often poorer** than if formulated as a tablet or capsule, (particularly if they are susceptible to hydrolysis)
- ✓ May support microbial growth and hence require incorporation of **preservatives**
- ✓ Bad taste of drugs can be felt
- ✓ Has less accuracy of dosing
- ✓ **Short shelf life**

Classification of solution dosage forms

Classified according to the route of administration

Oral

- Syrups
- Linctuses
- Elixirs
- Spirits

In mouth & throat

- Mouth washes
- Gargles
- Throat paint
- Throat sprays

Classification of solution dosage forms

Instilled into the body cavities

- Douches
- Enemas
- Ear drops,
- Nasal sprays
- On body Surfaces/topical
 - Collodions
 - Liniments
 - Lotions

1. Solutions taken orally

- Oral solutions are usually formulated so that the patient receives the usual dose of the medicament in a conveniently administered volume, usually a multiple of 5 mL.

These includes

- **Syrups**
- **Linctuses**
- **Elixirs**
- **Spirits**

Syrups

Syrups are concentrated aqueous preparations of a sugar or sugar substitute with or without flavoring agents and medicinal substances.

- Antihistamine Syrup
- Acetaminophen Syrup
- Cough and Cold Syrup



Syrups.....

Advantages of syrup

- Taste masking of disagreeable-tasting drugs.
- Provide a mean for extemporaneous compounding of medications that comes in tablets and capsules for those who can't swallow such dosage forms.
- Very appealing to children therefore, it useful for administration of medications to children.
- In the antitussive syrup preparations: the thick, sweet syrup will also have a **soothing effect on the irritated tissues of the throat as it passes over them.**

Syrups....

Components of syrup

- In addition to purified water and medicinal agent, syrup contains:
 1. Sugar or sugar substitute: provide sweetness and viscosity
 2. Preservatives
 3. flavorants
 4. Colorants

It may also contain special solvents, solubilizing agents, thickeners and stabilizers.

Syrups.....

- Sucrose is the most frequently employed sugar in syrup.
- Sucrose can be replaced by whole or in part by either:
 - **Other glycogenetic substances** (materials that convert to glucose in the body): Sorbitol, glycerin and propylene glycol.
 - **Non-glycogenetic substances**: methyl cellulose and ethyl cellulose.

Syrups....

- Non-glycogenetic substances are not hydrolyzed nor absorbed in the body.
- It provide and excellent syrup-like vehicle for administration of medication intended for diabetic patients or those on controlled or restricted to non-glycogenetic diet.
- Such artificial syrups contain artificial sweetener to produce similar syrup-like taste.

Syrups.....

- In theory, syrup NF (85 % w/v sucrose) requires no preservatives.
- The inherent stability of syrup is due to the unavailability of water required for microorganisms growth.
- However, preservatives are added when syrup is intended to be stored.

Syrups.....

- Saturated syrups are not recommended since some sucrose might crystallize out at low temperatures which, acting as crystallization nuclei, will result in separation of a considerable amount of sucrose from the syrup solution
- this will dilute the syrup resulting in susceptibility for microorganism growth.
- Most syrups contain 60-80 % w/v of sucrose:
 - Desirable sweetness and viscosity
 - Stability: resistance to microbial and mold growth.

Syrups.....

Preparation of syrups:

1. Solution with the aid of heat(Hot process)
2. Solution with agitation without the aid of heat(cold process)
3. Addition of sucrose to a medicated liquid or a flavored liquid

Syrups.....

1. Solution with the aid of heat

- Fast
- Volatile substances (alcohol, flavoring oils) or thermo-sensitive substances may be added after cooling of the hot solution.
- Over heating will result in formation of inverted sugar altered (sweeter) taste, darker color and more susceptible for microorganism growth.
- Excess over-heating will result in caramelization.

Syrups....

2. **Solution with agitation without the aid of heat:**
 - Advantage: maximum stability as it avoids heat conversion of sucrose.
 - Time consuming.
- **when simple or non-medicated syrup are employed as a sweetening and a vehicle**, its better to dissolve solid materials in minimal amount of purified water first, then add the resulting solution to syrup
- limited amount of available water in concentrated syrup.

Syrups....

3. **Addition of sucrose to a medicated liquid or flavored liquid**
 - Add sucrose to the aqueous solution of medication to make a syrup.

Linctuses

- A linctuse is a viscous liquid for oral use, the majority being for the relief of cough.
- The viscous nature of the preparation coats the throat & helps to alleviate the irritation which is causing the problem.
- Previously many linctuses contained a high level of sucrose; however many have been reformulated as 'sugar-free' products to reduce the risk of dental carries.
- Because the viscous nature of linctuses is beneficial they should not be diluted prior to administration e.g. dimorphine linctuses

Elixirs

- Which are clear sweetened **hydroalcoholic solutions** intended for oral use and
- **The proportion of alcohol present in elixirs varies widely** since the individual components of the elixirs have **different water and alcohol solubility characteristics** (each elixir require a specific blend of alcohol and water to maintain all of components in solution).



Elixirs

- In addition of alcohol and water other ingredients may be used in elixirs such as :
 - Glycerin and propylene glycol which used as **adjunct solvents**.
 - Sucrose , saccharine, and sorbitol as **sweetening agent**
 - **Flavouring materials** to increase their palatability.
 - **colouring agents** to enhance their appearance.
 - Elixirs containing over 10-12% of alcohol are usually self preserving .
 - Elixirs should be stored in tight ,light resistant containers and protected from excessive heat because of their usual content of volatile oil and alcohol.

Elixirs

- Non medicated elixirs are employed as **vehicles** and medicated elixirs are used for the **therapeutic effect** of the medicinal substances it contain.
- The therapeutic action of drugs presented as elixirs varies widely and includes **antihistamines, antibiotics and decongestants**.
- E.g. phenobarbital elixirs.

Preparation of elixir

- Elixirs are usually prepared **by simple solution method with agitation** and / or by the admixture of two or more liquid ingredients.
- **Alcohol soluble** and **water soluble** components are generally dissolved separately in alcohol and purified water, respectively.
- **Then the aqueous solution is added to alcoholic solution** rather than the reverse, in order to maintain the highest possible alcoholic strength at all times so that minimal separation of alcohol-soluble components occurs.
- When the two solutions are completely mixed the mixture is made to volume with specified solvent or vehicle.

Preparation of elixir

- In preparation of elixirs frequently the final mixture will not be clear (**cloud**) due to separation of some flavouring oils by reduced alcoholic concentrations.
- If this occurs the elixir is usually permitted to **stand for a prescribed number of hours** , to ensure the saturation of hydro-alcoholic solvents and to permit the oil globules to coalesce so that they are more easily removed by **filtration**.

Phenobarbital elixir U.S.P

Rx

Phenobarbital	4 g
orange peel	30 ml
Solution of amaranth	10 ml
Alcohol	125 ml
Glycerin	450 ml
Syrup	250 ml
D.W Q.S	1000 ml

Method:

1. Dissolve the phenobarbital in alcohol
 2. Add the tincture of orange peel ,glycerin, syrup, amaranth solution and add sufficient water to produce 1000 ml ,mix well and filter.
- The phenobarbital elixir used as sedative and hypnotic.
 - Orange peel used as flavouring agent.
 - Solution glycerin used as thickening agent ,also increase the solubility of phenobarb .
- Ammananth solution used as colouring agent. Syrup used as sweetening agent.

Differences between elixirs and syrups

1. Elixirs are usually less sweet and less viscous than syrups because they contain a lower proportion of sugar and consequently are less effective than syrups in masking the taste of medicinal substances.
2. Elixirs are better able than aqueous syrups to maintain both water soluble and alcohol soluble components in solution because of their hydroalcoholic character.
3. From a manufacturing standpoint, elixirs are preferred over syrup due to their stability and ease of preparation (by simple solution).

Spirits

- Are alcoholic or hydro alcoholic solutions of volatile principle (mostly volatile oil).

spirits can be classified according to their uses:

- ✓ **Therapeutic spirits:** which contain therapeutic volatile substances.
- ✓ **Flavoring spirits :** which contain flavoring volatile substances.
- spirits also known as **essences**.
- The amount of volatile substance in spirits varies greatly and no fixed percentage can be given ,**but in all cases, volatile substances in the spirits is more than that of aromatic water.**

Spirits

- Amount of alcohol differ from one spirit to another,
- the lowest amount of alcohol found in **internal spirits** as(aromatic spirit of ammonia) which contain 62-68% alcohol
- while (camphor spirit) which is used **externally** contain 80-87%.
- Spirits when taken orally , they are generally mixed with portion of water to reduce from pungency.

Spirits

- In preparation of spirits we must keep in mind that the oil dissolved in alcohol is precipitated causing **turbidity** when the solutions are mixed with water other than specified in the formula ,
- so all equipment should be dry and filter paper should be moisten with alcohol.
- Spirits should be stored in light resistant containers and in cool place to prevent evaporation of alcohol and volatilization of volatile substance or active ingredients .

Method of preparation of sprits

Simple Solution

- Majority of spirits are prepared by dissolving the solute in alcohol by agitation. Filtration is generally desirable to obtain a sparkling clear product.

Solution with Maceration

- Macerate the vegetable materials in a suitable solvent to remove the undesired constituents or to extract one which is desired.
- Example: Peppermint Spirit

Use: digestive aid or carminative

Comparison between elixirs and spirits

Elixirs

1. Contain sweetening agent
2. More viscous
3. Need preservative if the alcohol content is less than 10-12%
4. Water and alcohol are usually the primary solvent
5. For internal use only
6. Concentration of alcohol is lower

Spirits

1. doesn't contain
2. less viscous
3. no need
4. alcohol is usually the primary
5. used internally and externally
6. alcohol is 62-87%

Aromatic waters

- Aromatic waters are clear, aqueous solutions saturated with volatile oils or other aromatic or volatile substances(camphor).
- An aromatic water is water enriched with both the essential oil and the water-soluble volatile components of a plant. This contributes to their efficacy.
- Naturally, the odors and tastes of aromatic waters are of the volatile substances from which they are prepared.
- used for perfuming and/or flavoring
- Eg. Rose water, chloroform water, peppermint water

Aromatic waters

- Volatile oils solutions represent an incompatibility problem of **salting out**.
- This occurs after the incorporation of a very soluble salt in their solution. Therefore sometimes talc is added as a distributing agent.

Aromatic water deteriorates with time therefore:

- should be made in small quantities.
- Should be protected from intense light and excessive heat by storing in air tight, light resistant containers.
- should be discarded, if it become cloudy or otherwise deteriorate.
- Deterioration may be due to volatilization, decomposition or mould growth.

Aromatic Waters method of preparation

A .Distillation process

- The drug/plant is coarsely ground and mixed with sufficient quantity of purified water in the distillation unit.
- After distillation any excess oil in the distillate is removed by filtration.
- Advantage.: most satisfactory method.
- Disadvantage.: slow and expensive

Aromatic Waters method of preparation

B .Solution process

- Aromatic water may be prepared by shaking volatile substance with purified water for 15 minutes.
- The mixture is set aside for 12 hours & filtered.

Disadvantage

- *It is difficult to get clear solution due to the formation of extremely fine particles.*

Aromatic Waters method of preparation

3. Alternative solution Method:



By this method the volatile oil or suitably comminuted aromatic solid is thoroughly incorporated with 15gms of talc and to this mixture is added 1000ml of purified water. The resulting slurry is thoroughly agitated several times for the period of 30 minutes and then filtered.

Tinctures

Tinctures are alcoholic or hydroalcoholic solutions prepared from vegetable materials or from chemical substances.

It vary in

- ✓ Method of preparation,
- ✓ Strength of the active ingredient,
- ✓ Alcoholic content, and
- ✓ Intended use.

When it is prepared from chemical substances (e.g., iodine, thimerosal), tinctures are prepared by **simple solution of the chemical agent in the solvent.**

Tinctures

- Depending on the preparation, tinctures contain alcohol in amounts ranging from approximately **15% to 80%**.

Alcohol has two functions which include:

- ✓ Protects against microbial growth.
 - ✓ Keeps the alcohol- soluble extractives in solution
-
- Avoid mixing with liquids too diverse in solvent character, e.g. water since there will be precipitation of ingredients upon addition of water.

Tinctures

- Because of the alcoholic content, tinctures must be tightly stoppered and not exposed to excessive temperatures.
- Also, because many of the constituents found in tinctures undergo a photochemical change upon exposure to light, many tinctures must be stored in light-resistant containers and protected from sunlight.
- As most tinctures are not intended for oral use, advise the patient for the proper use. The label should state **for topical use**.

Tinctures

E g. Iodine Tincture

- Preparation:
 - Dissolve iodine crystals and potassium iodide in purified water then add alcohol qs. to produce hydroalcoholic solution.
- **Uses: local anti-infective.**

2. Solutions used in the mouth & throat

Gargles

- Are aqueous solution intended for use in the prevention and /or treatment of throat infection by forcing air from the lung through the gargle.
- Usually contain a bactericide.eg phenol or thymol.
- Eg. phenol gargle

Mouth wash

- Are aqueous preparation used to cleanse and deodorize the buccal cavity for the purpose of oral hygiene and to treat infection of the mouth. Eg. compound sodium chloride mouth wash

2. Solutions used in the mouth & throat

Mouth washes & Gargles

- These are aqueous solutions for the prevention & treatment of mouth & throat infections.
- Both are concentrated solutions although gargles tend to contain higher concentrations of the active ingredient than mouth washes.
- Both are usually diluted with warm water before use.
- Both contain antiseptics and weak astringents.
- The liquid is usually not intended for swallowing.

3. Solutions instilled into body cavities

Ear preparations

- Also known as otic or aural products,
- These are simple solutions of drugs in either water, glycerol, propylene glycol or alcohol/water mixtures for local use, and include antibiotics, antiseptics, cleansing solutions and wax softeners.
- They are applied to the external auditory canal as drops, sprays or washes.

3. Solutions instilled into body cavities.....

Enemas

- Enema is an oily or aqueous solution which is administered rectally.
- A variety of drugs are formulated as enemas & are used to treat conditions such as constipation or ulcerative colitis.
- They are also used prior to examination of the lower bowel and for systemic effects.
- **Eg. sodium phosphate Enema**

3. Solutions instilled into body cavities....

Vaginal douches

- Is an aqueous solution used for rinsing a body cavity and function as cleansing agent, antiseptic or astringent
- Vaginal douches are often dispensed in the form of powder with directions in specified quantity of water (usually warm).
- However, tablets for preparation of solution are available or the solution can be prepared by the pharmacist.
- Vaginal douches are used **for cleansing vagina.**
- **Eg. Potassium permanganate douche**

4. Topical solutions

▪ Liniments

- Liniments are **alcoholic** or **oil-based** solutions or emulsions containing therapeutic agents intended for external application.
- Liniments usually are applied with friction and rubbing of the skin, the oil or soap base providing for ease of application or massage.
- Alcoholic liniments are used generally for their rubefacient, counter-irritant, mildly astringent effects.
- The oily liniments are milder in their action are more useful when massage is required.
- Liniments should not be applied to broken or bruised skin.

4. Topical solutions.....

Paints

- Liquids for application to the skin or mucous membranes in small amounts are often termed **paints** and are usually applied with a small brush.
- A viscosity modifier such as glycerol is often added to ensure prolonged contact with the skin.

4. Topical solutions.....

- **Collodions** are similar preparations which, after evaporation of the solvent leave a **tough flexible film** that will seal small cuts or hold a drug in intimate contact with the skin.
- The solvent is normally alcohol, acetone or ether which evaporates quickly leaving a film on the skin that contains the active agent.
- The film former is usually pyroxylin (nitrocellulose) in an alcohol/ ether or alcohol/ acetone solvent blend.
- Often a plasticizer such as castor oil and an **adherent** such as colophony resin are included.

5. Injectables (sterile products)

- Are sterile solutions for injection into the body
- These are subdivided into:
 - **small volume parenteral fluids**
 - **large volume parenteral fluids**
- Small volume parenterals are sterile, pyrogen free injectable products. They are packed in volumes up to 100ml.
- Small volume parenterals are packed as:
 - **Single dose ampoules**
 - **Multiple dose vials**
 - **Prefilled syringes**

General method of preparation of solutions

- Most solutions are prepared by **simply mixing** of the solutes with the solvent.
- For both small and large scale manufacture of solutions the only equipment necessary is suitable **mixing vessels**, a **means of agitation** & a **filtration system** to ensure clarity of the final solution.

General method of preparation ...

- During preparation, the solute is simply added to the solvent in **a mixing vessel** & **stirring** is continued until dissolution is complete

General method of preparation

- If the solute is more soluble at elevated temperature, it may be advantageous to apply **heat** to the vessel particularly if the dissolution rate is normally slow.
- **Size reduction** of solid materials to increase their surface area also speed up the process of solution.

Methods to speed up dissolution of the solute

- Applying heat
- Reducing the particle size of the solute
- Using a solubilizing agent
- Stirring